

SMART PUBLIC DISTRIBUTION SYSTEM

ABSTRACT

Smart Public Distribution System (SPDS) is a web-based application designed to improve the efficiency, transparency, and accessibility of the Public Distribution System (PDS). The system enables ration card holders to check the availability of their monthly ration, view the operational status and working hours of nearby ration shops, receive notifications when monthly stock is available, and submit complaints or feedback regarding services.

The application consists of two major modules: **User Module** and **Admin Module**. Users can register and log in to access details of their ration shop, check the availability of essential commodities such as rice, wheat, sugar, and kerosene, view their monthly entitlement, and receive timely notifications. Administrators can manage ration shop information, update stock availability, maintain user records, respond to complaints, and generate reports for effective monitoring.

The primary objective of this system is to reduce unnecessary visits to ration shops, improve communication between beneficiaries and administrators, and enhance transparency in the ration distribution process. It also provides a centralized platform for stock management, complaint handling, and customer feedback, making the Public Distribution System more organized and efficient.

The proposed system is developed using modern web technologies with a responsive user interface and a secure database for storing user, shop, and stock information. Future enhancements may include integration with government PDS databases, SMS notifications, QR code authentication, mobile application support, and AI-based demand prediction for efficient stock management.

In conclusion, the **Smart Public Distribution System** offers a practical, user-friendly, and scalable solution for digitizing ration distribution services. The system improves convenience for beneficiaries, simplifies administration, and contributes to the digital transformation of public welfare services.